

Round 4 — Builder Feedback

Feedback on the DreamDEX protocol — mechanics, execution, and docs — from building and running an autonomous trading agent through the Round 4 volume contest.

1,501,818

USDSO VOLUME

~\$130k/hr

PEAK THROUGHPUT

~3×

GAS EFFICIENCY VS 2-TX

The round in one paragraph

Reached **1,501,818 USDso** of volume. The engineering focus was an **atomic (EIP-7702) taker** that places a buy and a sell in a single transaction, plus **depth-aware leg sizing** that fits each round-trip to the top-of-book depth. With zero protocol fees, the only costs are the **spread crossed** (the toll) and **gas** — the toll is ~fixed per \$1k of volume, gas is ~fixed per transaction. So the efficient move is fewer, larger transactions on the deepest book. On a deep book the atomic engine sustained **~\$130k/hr** at close to the spread floor.

What worked well

- **EIP-7702 atomic execution** is the standout. Bundling buy+sell into one transaction removed leftover-inventory risk entirely and roughly **tripled gas efficiency** versus two separate transactions — a great primitive on an L1 CLOB, making round-trip strategies both cheaper and safer.
- **Zero protocol fees** make the cost model clean: spread plus gas, nothing hidden. It rewards good execution rather than fee arbitrage.
- **Deep books absorbed large legs** with negligible price impact — \$198 legs on the deep pair moved at ~the quoted spread.

Protocol notes & suggestions

1. Collateral is reserved at the limit price, not mid

A buy reserves collateral sized at the **limit** price (above mid by the slip), so a leg whose notional equals the free balance pre-reverts on the very first order. We cap legs at ~0.95× free balance to avoid it. Defensible behavior, but not obvious from the docs — a one-line note (“orders reserve collateral at the limit price, not mid”) would save integrators a debugging session.

2. Per-pair book depth varies wildly

Top-of-book depth on one pair was ~\$8k while another pair’s bid side was ~\$100 at the same instant. Depth-aware sizing handles this, but a naive fixed-leg taker will silently walk the thin

book and pay far more than the quoted spread. Not a bug — just worth surfacing that per-pair depth is highly uneven, so integrators size to the touch rather than to a fixed notional.

3. Endpoint availability

During the round, every public Somnia RPC endpoint we use went unreachable at the same time for a stretch (our failover cycled through all of them and exhausted). Our engine handled it — it self-stops cleanly and flattens — but a synchronized outage is a real availability risk for anything automated against the exchange. Documented rate limits and at least one more independent endpoint would reduce the blast radius.

Carry-over from Round 1 findings

The protocol findings filed in Round 1 (`reports/FINDINGS.md`) still make useful regression targets — a few were confirmed and fixed by the team. The pre-trade `eth_call` false-negative rate and the `OrderPlaced filled=0` event semantics are the two most worth double-checking still hold after recent changes.